

Intelligent Traffic Preemption System for Emergency Vehicles: State of the Art, Gap Analysis, and IDP Engineering Blueprint

Executive Summary

Emergency Vehicle Preemption (EVP) has evolved through three broad generations: **optical line-of-sight systems**, such as Opticom IR and TOMAR STROBECOM; **GPS/cellular/cloud systems**, such as Miovision Opticom Cloud/GPS, Applied Information Glance, LYT.emergency, and Econolite Centracs Priority; and **standards-based connected-intersection systems** built around V2X messages such as SAE J2735 SRM/SSM and SAE J2945/B. The newest SAE J2945/B standard, issued in November 2025, is especially important because it formalizes an interoperable over-the-air mechanism for connected vehicles to request traffic-signal priority or preemption using Signal Request Messages and receive status through Signal Status Messages. ¹

The global market shows that **the basic problem of making an approaching ambulance or fire engine receive green has already been solved commercially**. Miovision Opticom supports IR, GPS, and cloud-based preemption; Applied Information combines GPS, LTE, 900-MHz radio and C-V2X; LYT provides cloud-based predictive preemption; Econolite Centracs Priority combines GPS/AVL/CAD data with ETA-based control; Q-Free performs V2X processing locally on ATC controllers; TOMAR provides mature optical systems; and Queensland operates a large centralized GPS/CAD-driven EVP network. ²

Therefore, an IDP should **not** be framed as merely:

“Detect an ambulance and make its traffic light green.”

That has been done for decades. A much stronger project is:

A standards-ready, authenticated, queue-aware, predictive, hybrid edge-cloud emergency corridor system that works with both legacy signals and future V2X infrastructure while minimizing disruption to normal traffic.

I recommend building a project provisionally called **PreemptEdge** or **RESQ-Green** with six distinguishing properties:

1. **Predictive, just-in-time preemption rather than distance-triggered preemption.** The system estimates emergency-vehicle ETA *and* the time required to discharge the queue ahead, starting clearance only when needed. Queue-discharge research has shown why traffic-dependent activation is preferable to a fixed trigger distance. ³

2. **Edge safety with cloud coordination.** Route computation and multi-intersection orchestration can occur centrally, but the intersection gateway independently validates every request, enforces timeouts, and never permits unsafe phase transitions.
3. **Authenticated preemption.** A request is accepted only when device identity, emergency authorization, active dispatch status, lights/siren state, freshness, route plausibility, and geofence conditions agree. IEEE 1609.2 provides the relevant security architecture for signed V2X application messages. ⁴
4. **Vendor-neutral interoperability.** Internally use a common priority-request model and provide adapters for NTCIP 1202/1211, SAE J2735/J2945/B, and a legacy isolated preempt input. NTCIP 1211 explicitly addresses preferential treatment of emergency and other priority vehicles. ⁵
5. **AI is advisory, not the safety controller.** Machine learning predicts ETA, route, congestion, anomalous requests, and optimal recovery. A deterministic safety state machine controls whether a preemption may physically occur.
6. **Low-cost migration path.** Demonstrate first with LTE/GNSS and a small edge computer rather than requiring expensive C-V2X RSUs at every intersection. Add true J2735/C-V2X communication as a future-compatible second path.

The project should combine a **SUMO digital twin, physical vehicle node, edge intersection gateway, controller emulator, central coordination server, cybersecurity layer, and operations dashboard**. Eclipse SUMO is a mature microscopic traffic simulator, while SUMO-RL provides Gymnasium/PettingZoo interfaces if constrained reinforcement learning is investigated later. ⁶

Most importantly, **do not let your Raspberry Pi, Jetson, ESP32, computer-vision program, or RL agent directly drive real red/amber/green lamps on a public intersection**. For an academic prototype, use a signal emulator or physical model. A later municipal pilot should communicate only through an authorized traffic controller's standards-compliant interface or certified preemption input. FHWA's current U.S. traffic-control reference is the MUTCD 11th Edition with Revision 1, effective March 5, 2026. ⁷

Standards, Technology Landscape, and System Models

The terminology matters. **Traffic Signal Priority** normally modifies timing while trying to preserve normal coordination; **preemption** is stronger and temporarily takes control away from the normal timing process to serve a higher-priority event. Modern systems blur this distinction by providing configurable classes and conditional behavior, but an emergency vehicle should generally be modeled as a high-priority preemption request rather than ordinary transit priority. NTCIP 1211 specifically standardizes signal-control-and-prioritization management information for classes including emergency-service vehicles. ⁵

The standards stack your project should understand is:

Layer	Relevant standard / technology	Why it matters to the IDP
Traffic controller	NTCIP 1202	Common controller objects for actuated traffic-signal control; the southbound adapter should be designed around controller objects rather than lamp-level wiring.

Layer	Relevant standard / technology	Why it matters to the IDP
Priority/preemption management	NTCIP 1211	Defines management information for Signal Control and Prioritization and preferential treatment of vehicle classes. ⁵
V2X messages	SAE J2735	Defines the V2X message dictionary. The September 2024 revision is the current cited release found in this survey. ⁸
Priority over V2X	SAE J2945/B	The 2025 standard specifies the OTA connected-vehicle/intersection TSP/preemption interface using SRM, SSM and WSA. ⁹
V2X security	IEEE 1609.2-2025	Secure application/management message formats and security functions. ⁴
U.S. traffic-control rules	FHWA MUTCD	The 11th Edition with Revision 1 is the current FHWA publication as of 2026. ¹⁰
European V2X	ETSI C-ITS CAM/DENM, SRM/SSM-compatible signal-priority profiles	Useful for keeping the architecture region-neutral instead of hard-wiring only North American protocols.
Simulation	Eclipse SUMO	Mature open-source microscopic mobility simulation with pedestrian and large-network support. ¹¹
Communication simulation	VaN3Twin / ns-3	Simulates 802.11p, LTE-V2X and NR-V2X with SUMO and includes emergency-vehicle alert examples. ¹²

SAE J2945/B is particularly relevant to making your project look like a serious ITS project rather than a student IoT demo. It specifies an interoperable method for connected vehicles to request priority/preemption and identifies the updated **SRM**, **SSM**, and **WSA** mechanisms; SAE's corresponding ASN.1 package provides machine-readable message structures. ¹³

A useful conceptual message exchange is:

SRM — Signal Request Message:

"Authorized emergency vehicle X is approaching intersection Y from approach Z and requests priority/preemption."

SSM — Signal Status Message:

"Intersection Y has received the request; status is pending/accepted/servicing/rejected/etc."

SPaT — Signal Phase and Timing:

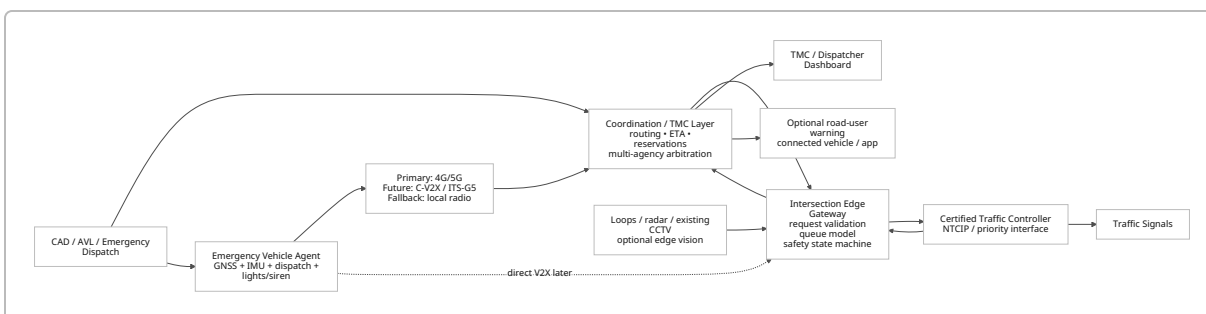
Current signal states/timing.

MAP:

Intersection geometry and lane connectivity.

Your own prototype does not need licensed SAE ASN.1 implementations immediately. You can initially create an internal message schema with equivalent concepts and construct a standards adapter later.

The most important architectural principle is to separate four planes:



This separation is close to where the industry is moving: cloud-centric products exploit fleet connectivity while controller-centric systems such as Q-Free MAXTIME cv process SRM, BSM and PSM locally and broadcast SPaT, MAP and SSM at the intersection. 14

Existing Commercial and Municipal Systems Worldwide

The table below focuses on systems that provide meaningful architectural lessons rather than listing every vendor.

System	Architecture and deployments	Strengths	Limitations / safety / privacy	Cost and complexity
Miovision Opticom	Supports Cloud, GPS and IR . Cloud uses existing integrations; GPS removes optical line-of-sight; IR uses vehicle emitter/ intersection detector. Its core sequence is detect → request → clear → resume. It supports relative priority and mutual-aid operation. ¹⁵	Extremely mature; gradual migration from optical to GPS/ cloud; broad ecosystem; hybrid operation avoids rip-and-replace.	Optical mode still needs line of sight and roadside detectors; GPS/ cloud modes introduce networking/ location-data dependencies. Proprietary ecosystem. Vendor performance figures should not be treated as universal independent results.	Medium-high at network scale. Historical San Marcos deployment: seven TxDOT intersections, \$77,000 in 2017, after which all city signals had EVP. ¹⁶
Applied Information Glance	Intersection FMU includes 4G LTE, GPS, 900-MHz radio , Ethernet and I/O; supports C-V2X network communication and a 5.9-GHz C-V2X RSU. ¹⁷	Strong communications redundancy; modern C-V2X pathway; remote health monitoring; useful for retrofit intersections.	Greater field-device complexity than pure-cloud products; LTE/ cloud service dependency; proprietary management platform and fleet provisioning.	Medium-high. Requires intersection FMU and possibly vehicle/C-V2X equipment; vendor quotes are deployment-specific.

System	Architecture and deployments	Strengths	Limitations / safety / privacy	Cost and complexity
LYT.emergency + Route Prediction	Software/cloud-first solution. LYT says route prediction learns historical emergency trips, continuously adjusts the predicted path and targets only signals on the expected route; earlier architecture used a secure edge device in the TMC to connect emergency vehicles to networked traffic signals. ¹⁸	Very low roadside hardware requirement when controllers are already networked; route prediction; TMC visibility; integrates with HAAS Alert/Whelen. ¹⁹	Requires a well-networked signal system and reliable backhaul/fleet positioning. Proprietary prediction logic must be treated as a safety advisory rather than assumed infallible.	Potentially low field CAPEX but recurring platform/integration cost; public unit pricing is generally not disclosed.
Econolite Centrac's Priority / Cobalt EOS	GPS/AVL/CAD-derived ETAs; Centrac's can plan several intersections ahead. EOS supports 16 preemption sequences , NTCIP-1211-based SCP for 10 priority classes , concurrent requests and a cabinet interlock. ²⁰	Excellent integration with real traffic-controller infrastructure; concurrency; controller-level interlocks; standards awareness; ETA-based operation.	Best fit where an Econolite/Centrac's ecosystem already exists; central software and CAD/AVL integration increase engineering complexity.	High integration complexity, quote-based. Richmond's 2025 project combines Centrac's Priority, CAD/AVL, TSP/EVP and predictive routing. ²¹

System	Architecture and deployments	Strengths	Limitations / safety / privacy	Cost and complexity
Q-Free MAXTIME cv / Dynamic CV	Runs directly on compliant ATC hardware. Broadcasts SPaT/ MAP/SSM; ingests SRM/BSM/PSM; supports emergency/transit priority without a separate central system. ¹⁴ Dynamic CV adds encrypted/ authenticated APIs. ²²	Closest commercial example to an edge-native, standards-centric architecture. Local operation reduces cloud latency and single-point dependency.	Requires compatible ATC controller and RSU/CV ecosystem to exploit all capabilities; deployment is less trivial on very old controllers.	Medium if modern ATC already installed; high if controller/RSU modernization is required.
TOMAR STROBECOM II	Vehicle-mounted coded infrared strobe → mast-arm optical detector → optical signal processor → controller preempt input. Detectors are mounted by traffic approaches. ²³	Deterministic and local; no cellular/ cloud dependency; 2,500+ ft rated optical range; self-diagnostics and automatic disable; confirmation-light option provides closed-loop visual acknowledgement. ²⁴	Requires line of sight and detector coverage for every relevant approach. Limited network-level route intelligence. Legacy coded optical identification is much less expressive than certificate-based V2X authorization.	Public 2026 emitter prices on TOMAR's site are roughly \$960-\$1,500 , before intersection detectors/ processors/ install. ²⁵

System	Architecture and deployments	Strengths	Limitations / safety / privacy	Cost and complexity
Queensland TMR Emergency Vehicle Priority	CAD + GPS + traffic-management information determine vehicle position/ETA; traffic signals along the route change when safe and return to normal after passage. Official TMR material reports more than 300 emergency vehicles and 1,100 signalized intersections across several Queensland regions at the cited program stage. ²⁶	One of the strongest large-scale examples of centralized route-aware EVP; integrates dispatch with traffic management rather than relying only on roadside sensing.	Central integration and fleet/controller coverage are substantial undertakings. The official page contains program-stage figures, not necessarily the complete 2026 installed base.	High institutional integration, but demonstrates city/state scalability.
Dutch intelligent-traffic-light ecosystem	Connected “intelligent traffic lights” recognize approaching emergency-service vehicles and provide priority while other directions are held; the broader Dutch approach combines traffic infrastructure with connected-vehicle information.	Particularly interesting for multi-agency and national interoperability , not just one vendor/city.	Requires coordinated infrastructure, telecom, credential and governance ecosystems; location data has stronger governance implications at national scale.	High ecosystem complexity, but potentially far lower marginal cost once intelligent signals are deployed broadly.

Several commercial products make performance claims—for example, Miovision advertises improvements in emergency response time and crash reductions—but these are vendor-reported aggregates and should **not** be used as your academic baseline without examining the underlying study design. ¹⁵ Your evaluation should instead compare identical simulated/field scenarios with preemption disabled versus enabled.

A few deployment cases are especially educational.

San Marcos, Texas demonstrates the economics of conventional retrofitting. Its official project page reports \$77,000 for Opticom installation across seven TxDOT intersections in 2017—about \$11,000 per intersection if naively divided, although that historical project-level ratio is not a current product quote. ¹⁶

Queensland demonstrates what a true institutional system looks like. Traffic control does not independently “recognize an ambulance”; emergency dispatch, GPS and traffic-management information collectively establish where the authorized responding vehicle is and when it will reach the signal. ²⁶ This is much harder to spoof than a simple siren detector or YOLO ambulance classifier.

Richmond, Virginia demonstrates the modern software approach: CAD/AVL integration, real-time location, route changes, ETA-driven priority and traffic-control coordination. ²¹

The clearest trend across these products is therefore:

optical detection → GPS detection → connected fleet/cloud → predictive route coordination → standards-based V2X + edge processing.

Your project should target the final two stages rather than reproduce the first two.

Academic State of the Art and High-Quality GitHub Projects

A rigorous IDP should distinguish academic algorithms from deployable safety systems. A paper may obtain excellent simulation performance yet have no mechanism for credentials, controller interlocks, pedestrian timing, radio failures, maintenance, certificate revocation or municipal integration.

The most relevant academic work is:

Work	Core idea	What you should reuse	What it still misses
Queue discharge-based EVP	Uses V2I plus queue size and a queue-discharge model to determine the optimal activation moment instead of triggering at a fixed distance; fine-tuned in SUMO and experimentally verified in urban traffic. ³	Best foundation for your MVP algorithm. Predict how long the queue needs to clear and activate just in time.	Primarily intersection/control logic; does not itself provide a full multi-agency cybersecurity/TMC product.
V2I kinematic-wave preemption	Uses shock-wave/kinematic-wave concepts to pre-clear background traffic at closely spaced intersections before the EV reaches them. ²⁷	Corridor-level predictive clearance and queue propagation.	Requires good traffic-state estimation; older communication assumptions.

Work	Core idea	What you should reuse	What it still misses
EMVLight	Joint dynamic routing + decentralized multi-agent RL signal control . The extended study reports up to 42.6% lower EMV travel time and 23.5% lower average trip time than its selected benchmarks in simulation. ²⁸	Route-control coupling, pressure-based reward, decentralized agents, emergency-capacity concept.	Primarily simulation evidence. It does not solve certification, real controller integration, communications security or fail-safe implementation.
EMGLight	DDPG jointly optimizes emergency priority and post-preemption recovery using a dynamic reward. ²⁹	Important insight: recovery is part of the problem , not an afterthought.	End-to-end continuous RL is difficult to safety-assure for a live traffic controller.
Communication-reliability-aware EVP research	Models the effect of V2X failures/interference rather than assuming perfect communication.	Gives a reason to maintain latency confidence, redundancy and fallback behavior.	Research models still need product-level health monitoring and certified fallback.

The academic literature points toward an important conclusion: the “smartest” algorithm is not necessarily the best real project. The strongest combination is:

Queue/traffic-model-based deterministic preemption + dynamic routing + constrained optimization + optional ML prediction + deterministic safety shield.

GitHub projects worth studying

I screened for projects with either institutional backing, mature adoption, significant engineering depth, or direct applicability. I would use the following as reference implementations.

Repository	Quality signal	What it actually provides	What to take into your project
USDOT/FHWA V2X-Hub	Official USDOT/FHWA project; mature plugin architecture and releases.	Infrastructure gateway translating/aggregating V2X and traffic-controller interfaces. Its plugin architecture includes SPaT/controller connectivity, priority-related processing and roadside communication.	Most important code reference for your interoperability/gateway layer. Copy the architectural idea of adapters/plugins rather than building one monolithic controller.

Repository	Quality signal	What it actually provides	What to take into your project
Eclipse SUMO	~4k stars and ~1.7k forks in the crawled GitHub snapshot; developed largely by DLR transportation researchers. ¹¹	Mature microscopic traffic simulator, large networks, multimodal traffic, pedestrians, route generation and TraCI control.	Your digital twin, baseline evaluation, queue measurement and hardware/software-in-loop foundation.
SUMO-RL	~1k stars and hundreds of forks; established research tooling. ³⁰	Gymnasium/PettingZoo environments on SUMO; lane density, queue observations; configurable rewards; Q-learning/PPO/DQN examples.	Use only for experimental recovery/coordination policies. Do not let an unconstrained RL action directly operate a physical signal.
VaN3Twin	Specialist academic V2X/ns-3 framework with implemented protocol stacks.	Integrates SUMO and ns-3; supports 802.11p, LTE-V2X Mode 4 and NR-V2X Mode 2; includes emergency-vehicle-alert examples in which emergency CAMs influence surrounding vehicles. ¹²	Excellent Phase-II environment for testing packet loss, radio congestion, latency, C-V2X and “vehicles clear the lane” behavior.
USDOT CARMA / CDA simulation ecosystem	Government-supported cooperative-driving stack.	Vehicle-side/cooperative-driving and simulation components for V2X applications.	Architectural ideas for standardized vehicle/infrastructure services and future cooperative emergency corridors.

A useful stack is therefore:

```

Eclipse SUMO
  ↓ TraCI
Your preemption engine
  ↑
SUMO-RL (only for experimental learning)
  ↑
VaN3Twin / ns-3 (communication realism)
  ↑
V2X-Hub-style protocol adapters

```

GitHub repositories I would *not* use as evidence

Many newly uploaded “smart traffic / ambulance priority AI” repositories consist of a YOLO notebook, Arduino LEDs, a README claiming an AI smart city, or a single SUMO scenario. Those can be useful for syntax but should not appear in your literature review as comparable systems.

For example, `sidd2305/Traffic_RL_Synapse` describes Q-learning in SUMO, a Pygame round-robin emergency-priority demo and heat-map visualization, but its repository metadata shows a very short April 2024 activity window and only a handful of stars/forks. It is a student concept, not an architectural reference for a deployable EVP system.

Likewise, `emceeashish/RL-Traffic-Signal-Control` has an unusually detailed README describing PPO, SUMO-RL, emergency flags and a 4x4 shared-policy grid, but the repository metadata shows it was created on February 28, 2026 with zero stars/forks in the captured metadata and only a very small commit history at that point. Its README is useful as an example of experiments to reproduce, but it is **not mature enough to treat its claimed performance as independent evidence**. Its README describes lane density, queues, emergency flags, PPO and multi-agent SUMO experimentation; that is interesting conceptually, but it needs reproducibility and independent validation before becoming an evidence source.

This filtering is important. **A beautiful dashboard, YOLO bounding boxes, or an RL reward plot is not evidence that a system can safely preempt a real traffic controller.**

What Current Systems Miss and What Your Project Should Build

The opportunity is not one single missing feature; it is the **combination** of features that commercial products, academic papers and typical student projects rarely provide together.

Current problem	Why it matters	Proposed technical solution
Fixed-distance activation	A 300-m trigger can be too early with no queue and too late with a long queue.	Queue-aware just-in-time trigger based on ETA, estimated queue discharge time and controller transition requirements.
Optical line-of-sight failure	Curves, trucks, weather, detector orientation and intersection geometry can prevent detection.	LTE/5G GNSS as primary MVP path; future C-V2X; optional local RF backup. Optical may remain as a compatibility input, not the sole channel.
Cloud outage / high latency	A centralized server must not become a safety-critical single point of failure.	Edge gateway caches policy and validates requests locally; heartbeat and watchdog; safe cancellation when commands expire.
Unauthorized or fake preemption	Simple RFID, siren detection or object detection can be spoofed or misclassified.	Hardware-backed certificates + dispatch binding + light/siren input + signed timestamp/nonce + geofence + map-matching + revocation. IEEE 1609.2 provides the relevant V2X security model. ⁴

Current problem	Why it matters	Proposed technical solution
Route changes trigger wrong intersections	Drivers may deviate around congestion or incidents.	Rolling corridor reservation; activate only the next few intersections; update route continuously and cancel old reservations.
Preemption harms cross traffic	Early activation can hold a corridor unnecessarily.	Predict queue-clearance time and trigger as late as safely possible, consistent with queue-discharge research. ³
Traffic remains disrupted after EV passes	Traditional systems often treat “vehicle passed” as the end of the control problem.	Recovery controller using queue debt / max-pressure logic to restore underserved approaches; EMGLight supports the importance of joint preemption/recovery optimization. ²⁹
Multiple emergency vehicles	Opposing ambulances/fire trucks can request incompatible greens.	Intersection reservation/arbitration engine with policy class, ETA, route conflict and request age; non-conflicting movements can potentially be served together.
Pedestrian clearance	Abrupt phase changes can create unacceptable risk.	Safety kernel never bypasses controller-configured minimum greens, pedestrian clearances, yellow and all-red transitions.
GPS errors	Urban canyons can shift the apparent approach lane.	GNSS + IMU + map matching; heading/lane confidence; reject automatic preemption when approach confidence drops below threshold.
Vendor lock-in	Cities contain mixed controller generations and vendors.	Southbound adapters: NTCIP 1202/1211, SAE J2735/J2945/B, and isolated legacy preempt input.
Poor maintenance visibility	A failed modem can remain unnoticed until an emergency.	Heartbeats, device inventory, signal/controller health, remote logs, signed OTA updates, A/B firmware rollback.
Privacy	Continuous first-responder tracking creates sensitive operational records.	Event-driven tracking only during authorized response; role-based access; short retention; aggregate long-term KPI data.
AI unpredictability	End-to-end RL can select unexpected actions outside training distribution.	AI predicts or recommends; deterministic safety state machine owns all signal transition authorization.

Proposed unique selling proposition

Your project can credibly claim:

“A cybersecure, predictive emergency green-corridor platform that calculates when—not merely whether—to preempt each intersection, uses edge safety enforcement,

supports legacy and V2X signal infrastructure, continuously reroutes, arbitrates multiple emergency vehicles, and actively repairs traffic disruption after the emergency vehicle passes.”

That is much stronger than “AI-based smart ambulance traffic light.”

The control algorithm

Let:

$$T_{EV,i}$$

be estimated emergency-vehicle time to intersection i .

Estimate:

$$T_{clear,i} = T_{startup} + \frac{Q_i}{s_i} + T_{transition} + T_{margin}$$

where:

- Q_i = vehicles estimated ahead of the EV or queue requiring discharge,
- s_i = estimated discharge/saturation service rate,
- $T_{startup}$ = startup lost time,
- $T_{transition}$ = time required by the legal/safe phase transition,
- T_{margin} = uncertainty margin based on positioning and network confidence.

A simple trigger is:

$$\text{PreparePreemption}_i = (T_{EV,i} \leq T_{clear,i}) \wedge \text{Authorized} \wedge \text{RouteConfirmed}$$

This is already more defensible than a fixed `if distance < 200 m: green()` rule because queue-discharge research specifically demonstrates adapting the activation point to actual queue conditions. 3

The system should choose among four strategies:

Green extension: EV is approaching a currently green compatible phase → extend it within permitted constraints.

Early green: incompatible phase is active → terminate it only after required safe timing, then service the EV approach.

Queue pre-clearance: significant queue lies ahead → begin serving that approach sufficiently early to empty it before arrival.

No intervention: EV will reach the intersection during an already favorable green or is too distant/uncertain to justify disruption.

That last option is important. An intelligent EVP is valuable partly because it knows **when not to preempt**.

Multiple emergency vehicles

Do not simply use “closest ambulance wins.” Use a policy hierarchy such as:

```
Infrastructure absolute safety override
    ↓
Jurisdiction-defined emergency class / incident urgency
    ↓
Confirmed active response
    ↓
Conflict/non-conflict geometry
    ↓
ETA to conflict zone
    ↓
Time already waiting for service
```

A future optimization layer can minimize:

$$J = w_e D_{EV} + w_n D_{normal} + w_q Q_{max} + w_s N_{switch} + w_r D_{recovery}$$

subject to hard signal-safety constraints.

AI/ML innovations that are actually worthwhile

ETA prediction: Gradient-boosted trees or a compact temporal model can learn segment travel time from time-of-day, queue state, speed, signal phase, road type and incident status.

Route prediction: Learn likely fire/EMS route choice from historical calls, similar in concept to commercial predictive systems such as LYT Route Prediction. ³¹

Vision-based queue measurement: A lightweight detector/tracker can estimate vehicle count, occupancy and queue tail from an existing camera. It should **not authenticate the ambulance**.

Anomaly detection: Flag impossible movement, duplicated vehicle credentials, repeated preemption without a dispatch event or contradictory GPS/IMU telemetry.

Constrained RL: In an advanced research extension, let RL choose *recovery timing or bounded timing adjustments*, never unrestricted signal indications. SUMO-RL makes this experimentation practical. ³⁰

Multi-agent coordination: EMVLight is useful inspiration for decentralized corridor cooperation, but implement deterministic baseline methods first and compare RL afterward rather than making RL a dependency. ²⁸

Complete Engineering Workflow, Hardware, Software, and Deployment Plan

The system should consist of five actual products developed together.

Emergency Vehicle Unit.

Provides trusted location and emergency status.

Intersection Edge Gateway.

Converts authenticated requests into safe requests to the existing controller.

Central Corridor Coordinator.

Maintains route, ETA, reservation and multi-agency state.

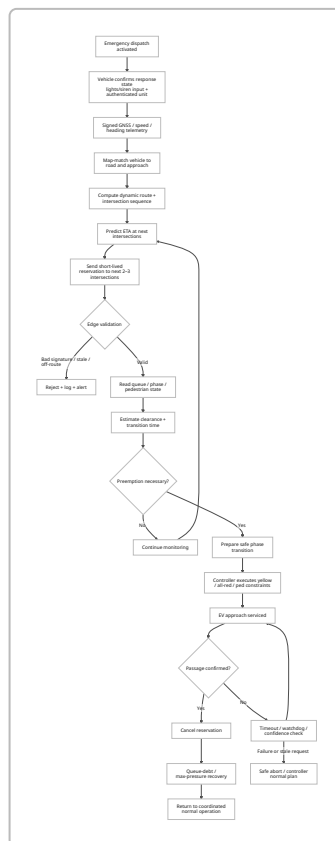
Digital Twin/Test Harness.

Reproduces the entire network in SUMO before hardware deployment.

TMC/Dispatcher Dashboard.

Lets authorized personnel see what the automatic system is doing and why.

The full event workflow should be:



Recommended prototype hardware

Component	IDP recommendation	Purpose
Vehicle processor	ESP32-S3 for minimal node, or Raspberry Pi Zero 2 W / CM4 for richer prototype	Telemetry, cryptographic signing, GNSS, status
GNSS	u-blox-class multi-constellation receiver	Position, velocity, UTC
IMU	Low-cost 6/9-axis IMU	Heading/dead reckoning and GPS sanity check
Cellular	SIMCom-class 4G module/router	Primary prototype communication
Secure identity	ATECC608-class secure element or TPM 2.0	Protect private device key
Emergency-state input	Opto-isolated 12/24-V input from a simulated light/siren circuit	Prevent “ordinary vehicle” requests
Edge gateway	Raspberry Pi 5 / CM4 for prototype; industrial fanless SBC later	Routing adapter, validation, queue/control logic
Vision accelerator	Optional Jetson Orin Nano / TPU-type accelerator	Only if doing camera-based queue estimation
Camera	Existing IP CCTV where possible	Queue/occupancy sensing
Controller interface	Ethernet/RS-232/RS-485 plus galvanically isolated digital I/O	NTCIP or laboratory preempt emulator
Physical demo signal	STM32/ESP32 + LEDs / scaled intersection	Safe hardware-in-the-loop demo
Backup power	UPS/DC backup for edge node	Fault-tolerance demonstration

For a real city deployment, the Pi/ESP32 versions are development devices. Replace roadside electronics with industrial-temperature, isolated, managed hardware and use the existing approved traffic controller as the final authority.

Recommended software stack

Layer	Technology
Simulation	Eclipse SUMO + TraCI/libsumo
RL experiments	SUMO-RL + Gymnasium/PettingZoo; Stable-Baselines3 or RLlib only for research
V2X simulation	ns-3 + VaN3Twin
Vehicle software	C++/Rust/Go on embedded Linux or C++ on ESP32

Layer	Technology
Edge safety/controller service	Rust or Go preferred for a robust service; Python acceptable for initial prototype
ML/analytics	Python, NumPy, scikit-learn, PyTorch if required
Backend API	FastAPI or Go
Event bus	MQTT initially; NATS/Kafka only if scaling requires it
GIS	PostgreSQL + PostGIS
Routing	OpenStreetMap + OSRM/Valhalla/GraphHopper
Cache/state	Redis optional
Dashboard	React + TypeScript + MapLibre GL
Metrics	Prometheus + Grafana
Logs	Loki/OpenSearch or structured PostgreSQL event logs for prototype
Identity	X.509 certificates + mTLS; Keycloak/OIDC for operator accounts
Packaging	Docker Compose first; Kubernetes/K3s only after it becomes useful

Avoid adding technologies merely to make the architecture diagram larger. An IDP with MQTT, Kafka, Kubernetes, blockchain, YOLO, LLMs and reinforcement learning simultaneously will probably be less convincing than a smaller system whose timing, safety and fault behavior you can prove.

Internal priority-request message

Your prototype can define:

```
{
  "request_id": "uuid",
  "vehicle_id": "pseudonymous-certified-id",
  "agency": "EMS",
  "response_class": "EMERGENCY_HIGH",
  "dispatch_event_id": "event-token",
  "latitude": 0.0,
  "longitude": 0.0,
  "speed_mps": 0.0,
  "heading_deg": 0.0,
  "lights_siren_active": true,
  "route_version": 14,
  "intersection_id": "I_042",
  "approach_id": "northbound_through",
  "eta_ms": 8200,
  "expires_at": "...",
}
```

```
"nonce": "...",  
"signature": "..."  
}
```

Later, the V2X adapter maps the relevant fields into the SRM/SSM workflow described by SAE J2945/B rather than forcing your entire application to depend on one transport protocol. ³²

Cybersecurity mechanism

A preemption request should pass approximately:

$$\text{Accept} = \text{CertValid} \wedge \text{DispatchValid} \wedge \text{ResponseActive} \wedge \text{Fresh} \wedge \text{GeoPlausible} \wedge \text{RoutePlausible} \wedge \text{PolicyAllow}$$

Use:

- per-device private keys stored in secure hardware;
- mutual TLS for LTE/cloud traffic;
- signed application payloads for commands;
- timestamps + nonces for replay protection;
- short reservation TTLs;
- credential revocation;
- RBAC for dispatch/TMC users;
- append-only signed security events;
- rate limits;
- signed OTA firmware;
- rollback-capable A/B update partitions.

IEEE 1609.2's secure message model provides an appropriate future V2X foundation rather than inventing unauthenticated "ambulance IDs." ⁴

Fail-safe mechanism

Every command needs a short validity period. The edge gateway should never receive:

```
TURN NORTH_SOUTH GREEN
```

It should receive something conceptually like:

```
AUTHORIZED PREEMPTION REQUEST  
approach = NORTHBOUND  
expires = t + 3 seconds  
priority_class = EMS_HIGH
```

The certified/simulated controller decides how to achieve that service while preserving transition constraints.

Failures should behave as follows:

Failure	Safe behavior
Cloud link lost	Existing local controller continues normal operation; active short-lived request expires unless edge can validate continuation locally.
Vehicle telemetry lost	Maintain only a bounded grace interval, then cancel.
GPS confidence poor	Stop issuing new automatic reservations; retain normal signal plan.
Edge gateway crashes	Controller remains independent and resumes/continues configured plan.
Camera fails	Fall back to detector-independent conservative timing; camera is not required for authentication.
Invalid credential	Reject; do not alter signal.
Conflicting emergency requests	Apply configured arbitration, not first-packet-wins.
Stuck request	Maximum preemption-duration watchdog triggers safe recovery.
ML process fails	Deterministic ETA/rule-based baseline continues.
Database unavailable	Local controller and edge safety remain operational.

That behavior differentiates a genuine ITS architecture from an IoT prototype.

Cost model

These are **engineering planning estimates for an academic prototype, not commercial quotations**.

Item	Prototype estimate
Emergency vehicle node: GNSS + 4G + secure element + enclosure/power	₹8,000–₹25,000
Higher-end Raspberry-Pi/industrial-style vehicle gateway	₹20,000–₹50,000
Intersection Pi/CM-class gateway + interfaces + power	₹15,000–₹40,000
More production-like industrial edge gateway	₹40,000–₹1,00,000+
Optional camera	₹5,000–₹20,000
Physical scaled signal demo	₹5,000–₹20,000
Small server/cloud prototype	roughly ₹2,000–₹15,000/month depending on architecture

Item	Prototype estimate
Four-intersection academic/HIL pilot	approximately ₹2–₹5 lakh excluding certified municipal controller/civil works
C-V2X RSUs	Do not include in MVP budget; add during standards/V2X phase

For perspective, conventional commercial systems are materially more expensive after certified roadside installation. San Marcos reported \$77,000 for seven Opticom intersection installations in 2017, while TOMAR currently publicly lists vehicle optical emitters alone around \$960–\$1,500 before intersection-side equipment and installation. ³³

Your cost advantage should therefore be described as:

“Uses existing networked controllers and commodity authenticated fleet connectivity wherever possible; adds an inexpensive edge gateway only where local processing or a legacy interface is needed.”

Do not claim that your ₹20,000 Raspberry Pi box is a direct replacement for a certified \$10,000 roadway system; environmental qualification, electrical isolation, support, installation, regulatory approval and lifecycle maintenance explain much of the difference.

IDP development timeline

Time	Build	Deliverable
Weeks 1–3	Requirements, intersection model, standards abstraction, threat model	System specification; state machine; use cases
Weeks 4–6	SUMO network + emergency routes + fixed-time baseline	Repeatable digital twin and baseline KPIs
Weeks 7–9	Vehicle telemetry simulator and physical GNSS/LTE node	Signed location/status messages
Weeks 10–12	Intersection edge service + controller emulator	Safe green-extension/early-green state machine
Weeks 13–15	Queue estimation + just-in-time algorithm	Predictive queue-clearing demonstration
Weeks 16–18	Multi-intersection rolling corridor + dynamic routing	Green corridor that adapts to route changes
Weeks 19–20	Multi-EV arbitration + recovery algorithm	Concurrent-emergency scenarios
Weeks 21–22	Security attacks, communication loss, GPS errors, watchdogs	Fault-injection and cybersecurity test report

Time	Build	Deliverable
Weeks 23–24	Hardware-in-loop model, dashboard, benchmarking	Final IDP demo + performance report

A later real deployment should proceed much more conservatively:

```

Digital twin
  ↓
Bench controller emulator
  ↓
Hardware-in-loop
  ↓
Closed-course / non-public signal
  ↓
Shadow mode at a real intersection
(no control, predictions only)
  ↓
Supervised single-intersection pilot
  ↓
4–10 intersection corridor
  ↓
Multi-agency network

```

Shadow mode is especially valuable. Install the system but initially let it only record “I would preempt now.” Compare its decisions against actual traffic without changing a single signal. That gives traffic engineers evidence before control authority is enabled.

Evaluation, Interoperability, Compliance, and Final Project Positioning

The project should be assessed in terms of **emergency performance, network impact, safety, communications, security, maintainability and cost**, not merely “ambulance reached faster.”

A strong KPI matrix is:

KPI	Measurement	Suggested IDP target
EV travel time	Dispatch-route travel time vs no-preemption baseline	≥20% reduction in selected congested simulation scenarios
EV intersection delay	Time lost per controlled intersection	Minimize; report mean + p95

KPI	Measurement	Suggested IDP target
Green-on-arrival rate	EV enters conflict zone on requested service phase	≥90% in controlled simulation
EV stops	Stops per emergency route	Significant reduction vs baseline
Preemption duration	Seconds of abnormal signal operation	Lower than fixed-distance preemption
Normal-traffic delay	Added average/p95 delay	Demonstrate bounded impact
Maximum side-road queue	p95 / maximum queue	Demonstrate recovery controller advantage
Network recovery time	Time to return near baseline queue state	Compare recovery algorithms
False activation	Unauthorized/incorrect accepted preemptions	0 in defined test suite
Attack rejection	Replayed/expired/bad-signature requests rejected	100% in defined security tests
Safety invariant	Conflicting-green / invalid transition event	0
Edge decision latency	Validation + local decision p95	Target <200 ms on local hardware
End-to-end command latency	Vehicle → system → edge	Record p50/p95/p99; design target <500 ms under lab network
Availability	Valid request-processing availability	>99% during controlled pilot
Route-change cancellation	Time to withdraw obsolete reservations	<1–2 telemetry cycles
Maintenance effort	intervention/device/month	Compare centralized vs edge configuration
Cost	hardware + installation per intersection/vehicle	Report separately rather than one blended number

Those percentages are **project targets**, not claims about guaranteed real-world emergency response.

Your experimental design should have at least these baselines:

- B0 Normal fixed/adaptive signals – no EVP
- B1 Fixed-distance preemption
- B2 ETA-only preemption
- B3 Queue-aware just-in-time preemption

B4 Queue-aware + rolling corridor
B5 B4 + intelligent recovery
B6 Optional constrained ML/RL enhancement

Run each under:

Low / medium / saturated traffic
One emergency vehicle
Two compatible emergency vehicles
Two conflicting emergency vehicles
Route deviation
Long queue
Pedestrian call active
Packet latency
Packet loss
Cloud failure
GPS error
Invalid/spoofed request
Intersection-edge reboot

That experiment matrix can become a very strong dissertation/results chapter.

For TMC integration, design clear interfaces:

```

                                NORTHBOUND
CAD / AVL -----+
ATMS / TMC -----+--> Coordinator
Analytics -----+
Operator identity -----+

                                SOUTHBOUND
Coordinator --> Edge --> NTCIP 1202/1211 --> Controller
                        \--> Legacy certified preempt interface
                        \--> J2735/J2945/B SRM/SSM adapter
```

The U.S. **V2X-Hub** architecture is worth studying specifically because it demonstrates the value of a plug-in translation/gateway layer between connected-vehicle messages and infrastructure systems rather than embedding protocol logic throughout the application.

For interoperability, define one internal object:

PriorityRequest
PriorityStatus
IntersectionState

VehicleState
DetectorState

Then implement adapters:

```
LTE/MQTT JSON  ┌
SAE J2735 SRM  ┤──> PriorityRequest ─> Decision Engine
CAD/AVL event  └

Decision Engine ─> PriorityStatus ┤──> MQTT/WebSocket
                                   ├──> J2735 SSM
                                   └──> TMC dashboard
```

This is a major USP because the **control algorithm does not care whether the request originated through cellular, C-V2X, an authorized CAD system or a future radio protocol.**

For U.S.-style deployments, the standards roadmap should include the current FHWA MUTCD, NTCIP controller/prioritization interfaces, SAE J2735/J2945/B for connected priority/preemption and IEEE 1609.2 for V2X security. FHWA identifies the December 2025 MUTCD 11th Edition with Revision 1 as its current official publication, with Revision 1 effective March 5, 2026. ⁷ SAE J2945/B was issued on November 20, 2025 specifically to support nationally interoperable connected-vehicle traffic signal priority/preemption. ⁹

For Australia, signal installations remain subject to Australian/state requirements; Queensland, for example, states that its MUTCD supplements/takes precedence as specified over relevant Australian Standard provisions. ³⁴ Queensland is also actively modernizing more than 1,400 intersection infrastructure components and explicitly cites preparation for future connected and automated vehicles, showing why controller modernization and EVP should be viewed as part of a broader ITS migration path. ³⁵

For an Indian implementation, SAE/NTCIP should be treated as **interoperability/reference engineering standards, not automatically as Indian legal requirements.** A public-road pilot would need authorization from the signal-owning road agency/ULB/traffic police, compatibility approval for the installed traffic-signal controller and cabinet, and applicable telecom/spectrum approval for whatever radio technology is used. This is another reason that LTE-based authenticated telemetry plus a controller emulator is preferable for the university phase: it lets you prove the system concept without illegally or unsafely manipulating public-road infrastructure.

The final feature matrix shows how the proposed project can differentiate itself:

Capability	Traditional optical	Cloud EVP	Typical academic RL project	Proposed PreemptEdge
Emergency preemption	✓	✓	✓	✓

Capability	Traditional optical	Cloud EVP	Typical academic RL project	Proposed PreemptEdge
No line of sight required	✗	✓	n/a	✓
Works if cloud unavailable	✓	Usually limited	Usually n/a	✓ edge fallback
Route prediction	✗	Some	Some	✓
Dynamic rerouting	✗	Some	EMVLight: ✓	✓
Queue-aware trigger timing	Usually ✗	Product-dependent	Some research	✓ core feature
Multi-intersection corridor	Limited	✓	✓	✓
Multiple EV arbitration	Limited	Product-dependent	Often weak	✓
Post-preemption recovery	Basic	Product-dependent	Some research	✓ explicit controller
Cryptographic authorization	Legacy-limited	Product-dependent	Usually ✗	✓ core feature
Dispatch verification	Usually ✗	Often ✓	Usually ✗	✓
GPS/IMU plausibility checks	✗	Product-dependent	Usually ✗	✓
NTCIP-ready	Controller dependent	Product-dependent	Usually ✗	✓ adapter
SAE J2735/J2945/B path	✗	Emerging	Rare	✓ planned adapter
Edge safety kernel	Local hardware	Product-dependent	Usually ✗	✓
AI isolated from safety	n/a	Unknown/proprietary	Usually ✗	✓ architectural rule
Existing CCTV reuse	✗	Product-dependent	Often camera only	✓ optional queue sensing
Low-cost legacy retrofit	Medium	Depends on signal network	Prototype only	✓ explicit objective
Offline controller autonomy	✓	Varies	Rarely considered	✓

Capability	Traditional optical	Cloud EVP	Typical academic RL project	Proposed PreemptEdge
Signed OTA + rollback	n/a/varies	Varies	Rare	✓
Explainable operator dashboard	Limited	✓	Often demo UI	✓ decisions + reasons + health
Communication fault testing	Physical limitation	Proprietary	Rare	✓ ns-3/fault injection
Security attack testing	Rare	Proprietary	Rare	✓ KPI requirement
Open simulation/test harness	✗	✗	Often ✓	✓

The project should therefore be presented not as an attempt to beat Opticom, Glance, LYT or Econolite at their own mature commercial product, but as an **open research architecture that combines the strongest ideas of each generation**:

- from **TOMAR/Opticom optical systems**: deterministic local operation and positive confirmation; ³⁶
- from **Queensland**: dispatch + GPS + traffic-management integration; ²⁶
- from **Glance**: redundant communications and C-V2X evolution; ¹⁷
- from **LYT**: route prediction and targeted corridor activation; ³¹
- from **Centracs**: ETA-based planning, multiple priority classes and controller interlocking; ²⁰
- from **Q-Free**: edge processing and standards-based SRM/SSM/SPaT/MAP handling; ¹⁴
- from **queue-discharge research**: traffic-dependent activation timing; ³
- from **EMVLight**: coupling routing and signal control; ³⁷
- from **EMGLight**: treating traffic recovery as an optimization problem; ²⁹
- from **V2X-Hub, SUMO, SUMO-RL and Van3Twin**: open, reproducible infrastructure and simulation rather than an opaque demonstration. ³⁸

The strongest final IDP demonstration would therefore show, on one screen, an emergency vehicle moving through a SUMO or physical corridor while the dashboard displays:

AMBULANCE A12	
Authenticated	✓
Dispatch event	✓
Lights/siren	✓
GNSS/map-match confidence	97%
Current route	Hospital → Incident
Next intersection	I-04
ETA	12.6 s
Queue ahead	8 vehicles

Estimated clearance	7.4 s
Safe transition requirement	3.1 s
Decision	PRE-CLEAR IN 2.1 s
Request state	ACCEPTED
Communication	LTE primary / backup available

The corresponding intersection should display:

```

NORMAL
↓
VALIDATED REQUEST
↓
QUEUE CLEARANCE
↓
SAFE PREEMPTION
↓
EV PASSED
↓
QUEUE-DEBT RECOVERY
↓
NORMAL COORDINATION

```

Then deliberately disconnect the cloud, transmit an expired request, inject a fake vehicle credential, alter the emergency route, create two competing emergency vehicles, and generate a long cross-street queue. A project that can **demonstrate correct behavior under those failures** will be substantially more rigorous and distinctive than the large majority of “AI ambulance traffic light” projects on GitHub.

The academic contribution can consequently be framed precisely:

Design and experimental evaluation of a low-cost, standards-ready, cybersecure, predictive emergency-vehicle traffic-preemption architecture combining rolling route reservations, queue-discharge-based just-in-time activation, edge-enforced safety, dynamic route adaptation, multi-emergency arbitration and congestion-aware post-preemption recovery.

That scope is technically substantial, feasible as an IDP when implemented first through SUMO plus hardware-in-loop, extensible toward real C-V2X/J2945/B infrastructure, and differentiated from both low-level optical EVP and simulation-only AI traffic-control projects.

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